

# H200 vs RTX PRO 6000 Blackwell

LLM serving performance brief — Qwen3.8-27B and DeepSeek-V4-Flash-0731, measured on production-style multiturn workloads

Our position: NVFP4 is a weight-compression feature, but production serving is decided by memory capacity, memory bandwidth, and interconnect — and on all three the H200 leads: 141 GB HBM3e at 4.8 TB/s with up to 4-way NVLink, vs 96 GB GDDR7 at 1.8 TB/s on PCIe. Because KV caches stay FP8 on every production stack, NVFP4's lever covers only the weights — roughly half of device memory at high availability — and its measured benefit (their own A/B: Qwen3.8-27B, one RTX card) is +13% decode and −21% cold prefill. Real, but nowhere near the 2.7× hardware gap.

|                                | RTX PRO 6000 Blackwell                                    | NVIDIA H200                                                 |
|--------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|
| Memory / GPU                   | 96 GB GDDR7 · ~1.8 TB/s                                   | 141 GB HBM3e · 4.8 TB/s (2.7×)                              |
| Interconnect                   | PCIe Gen5; NVLink pairs only (TP ≤ 2) *                   | NVLink bridges, up to 4-way (TP ≤ 4)                        |
| KV cache                       | FP8 (typical, both systems)                               | FP8 (typical, both systems)                                 |
| Qwen3.8-27B decode, 1 GPU each | 66–75 t/s (FP8 → NVFP4)                                   | 138–234 t/s (2.1–3.1× faster)                               |
| DeepSeek-V4-Flash-0731         | 2× RTX · ≤32 users · 58 s cold TTFT at 32 users (32k ctx) | 4× H200 · ≤128 users · 19 s cold TTFT at 32 users (32k ctx) |

\* Server Edition has no NVLink at all; workstation cards may bridge pairs of two (TP ≤ 2). Anything larger crosses PCIe.

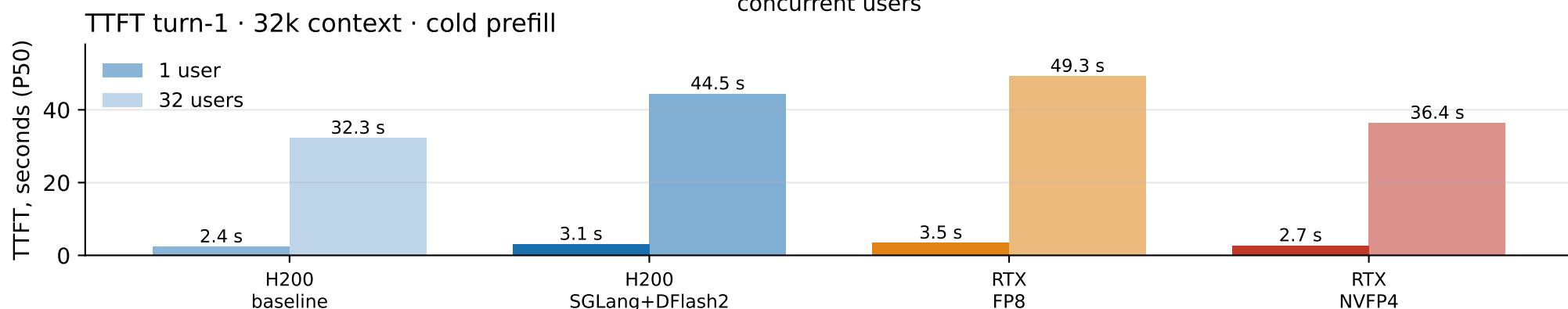
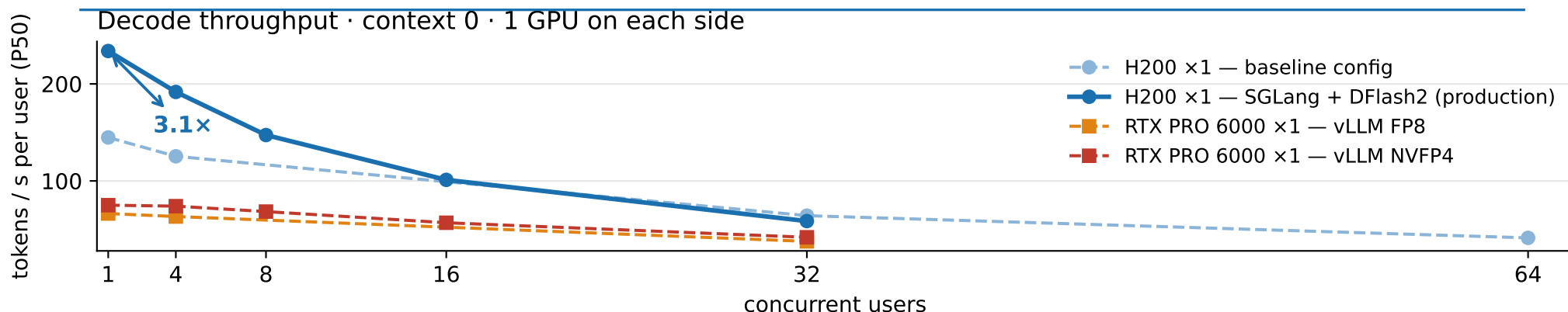
## Why this holds even under 4-bit / trellis quantization

Weight-only quantization is GPU-neutral: 4-bit expert weights already run on H200 (DeepSeek's MXFP4 experts via Marlin kernels), and trellis/QTIP-style quants are likewise weight-only, dequantized in-kernel — no Blackwell tensor cores required. Compressing weights shrinks both GPUs' byte streams equally, so the 2.7× bandwidth ratio and 1.5× capacity ratio survive any quantization level. KV caches stay FP8 throughout: the memory that scales with context length never benefits from weight formats.

Source: ModelBenchmarker results/ (multiturn benchmark: 5 turns/user, simultaneous users, tasks = coding/creative/mixed; plotted values are P50 means across tasks). RTX PRO 6000: 96 GB GDDR7, ~1.8 TB/s, PCIe (workstation pairs may bridge 2 GPUs; Server Edition has no NVLink). H200: 141 GB HBM3e, 4.8 TB/s, NVLink bridges up to 4-way (TP ≤ 4, 900 GB/s per-GPU aggregate).

1x RTX PRO 6000 (FP8 / NVFP4) vs 1x H200 — identical benchmark, P50 values averaged across tasks.

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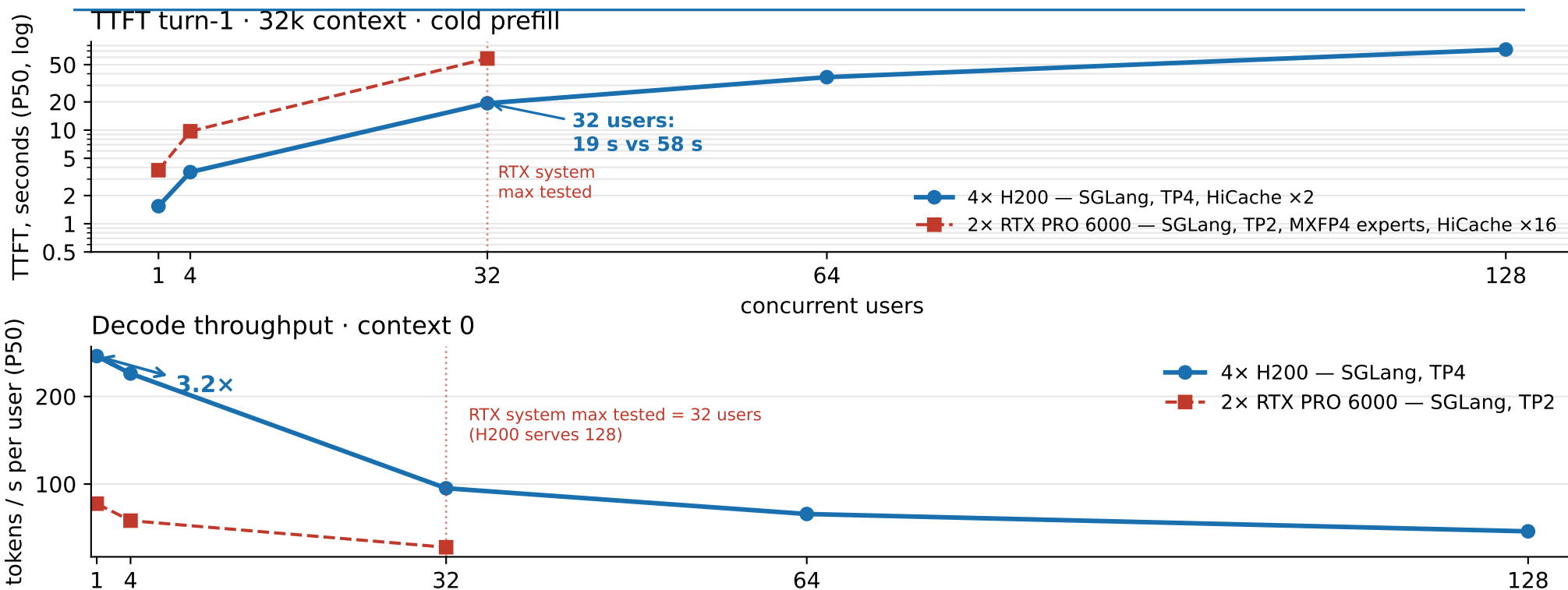


- Decode — the metric users feel — is HBM-bandwidth-bound: 1× H200 delivers 138 t/s baseline and 234 t/s with our production config (DFlash2), vs 66 (FP8) / 75 (NVFP4) t/s on the RTX — 2.1–3.1× faster per GPU.
- NVFP4's measured win on the RTX itself (their A/B): +13% decode, −21% cold TTFT (75 vs 66 t/s; 2.72 vs 3.45 s at 32k). Real, but nowhere near the 2.7× memory-bandwidth gap.
- At 32 users × 32k context, both RTX configs dropped 30+ requests (timeouts); the H200 dropped 0.
- Config note: the H200 DFlash2 series adds speculative decoding; RTX runs are stock vLLM. Even the plain H200 (no speculative decoding) doubles the RTX's best NVFP4 number.

***NVFP4 made the RTX PRO 6000's weights smaller. It did not make its memory faster, its cache larger, or its fabric wider.***

# DeepSeek-V4-Flash-0731 — 291B MoE · 4-bit expert weights on both systems

4× H200 vs 2× RTX PRO 6000 — the largest model that fits on each system, identical benchmark.



- Per-GPU decode: 246 vs 78 t/s single-user (3.2×); 95 vs 28 at 32 users. Decode is bandwidth-bound — the 2.7× HBM gap shows directly.
- Cold prefill at 32k / 32 users: 19.3 s (4× H200) vs 58.1 s (2× RTX). The H200 system serves 128 concurrent users; the RTX system was sized at 32.
- Warm (cached-prefix) latency is comparable (~0.5–0.8 s on both) — the systems part ways under load and long context, which is exactly where production traffic lives.
- The 2× RTX config only fits the model with 4-bit expert weights plus 16× host-RAM cache tiering; the H200 runs it in-device with 2× tiering. Weight-only quantization rescues the RTX's capacity — not its bandwidth or its fabric.

**Weights-only compression is GPU-neutral — it works on H200 today, and it compresses both sides equally. GPU choice is decided by memory, bandwidth, and interconnect.**

Source: ModelBenchmarker results/ (multiturn benchmark: 5 turns/user, simultaneous users, tasks = coding/creative/mixed; plotted values are P50 means across tasks). RTX PRO 6000: 96 GB GDDR7, ~1.8 TB/s, P